

From a Good Question to Reproducible Research

Week 4 • FINER, FAIR, provenance, metadata, and documentation



Today's question

What makes a biomedical study both worth doing and possible to reproduce?

Question

Is the research question worth asking?

Data

Can we obtain and understand appropriate data?

Process

Can another researcher follow what we did?

Result

Can the result be trusted and reused?

Learning objectives

- Use FINER to evaluate a biomedical research question.
- Distinguish FINER from FAIR and explain why both matter.
- Describe data provenance and why it supports reproducibility.
- Create a useful data dictionary and README.
- Document transformations from raw data to analysis-ready data.

Week 3 → Week 4

Version control records change; documentation explains meaning.

Week 3

Folders • raw vs processed •
Git • commits • branches

Week 4

Research question •
provenance • metadata • data
dictionary • README

Together

A traceable, understandable,
reproducible project

FINER: is this a good research question?

A framework for evaluating the question before investing in the study.



Feasible



Interesting



Novel



Ethical



Relevant

F — Feasible

Can we realistically answer the question?

- Enough participants or observations?
- Adequate statistical precision or power?
- Access to the required data or specimens?
- Enough time, funding, expertise, and computing resources?
- Can the variables actually measure the concepts in the question?

Interesting, Novel, Relevant — connected, but not identical

Interesting

Will investigators, participants, clinicians, or the scientific community care about the answer?

Novel

Does the question add new knowledge, a new population, method, comparison, or perspective?

Relevant

Could the answer matter for science, clinical care, policy, public health, or future research?

E — Ethical

A feasible study is not automatically an acceptable study.

- Are risks to participants justified and minimized?
- Are consent and privacy handled appropriately?
- Is the proposed use consistent with approvals and agreements?
- Are vulnerable populations treated appropriately?
- Could the analysis or reporting cause avoidable harm?

FINER in practice: a readmission study

Question: Among adults with type 2 diabetes, can admission information predict 30-day hospital readmission?

Feasible

Enough encounters? outcome frequency?
variables available?

Interesting

Would clinicians/data scientists want the
answer?

Novel

What does this add beyond existing models?

Ethical

Appropriate use of clinical data?

Relevant

Could it improve care or resource planning?

FINER ≠ FAIR

They answer different questions.

FINER

Is the research question worth pursuing?

Feasible • Interesting • Novel • Ethical • Relevant

FAIR

Are data and related resources managed so they can be found, accessed appropriately, combined, and reused?

Findable • Accessible • Interoperable • Reusable

FAIR principles

Good stewardship does not mean “make everything public.”

F — Findable

People can discover the data and metadata.

A — Accessible

There is a clear, appropriate route to access.

I — Interoperable

Formats and vocabularies support combination.

R — Reusable

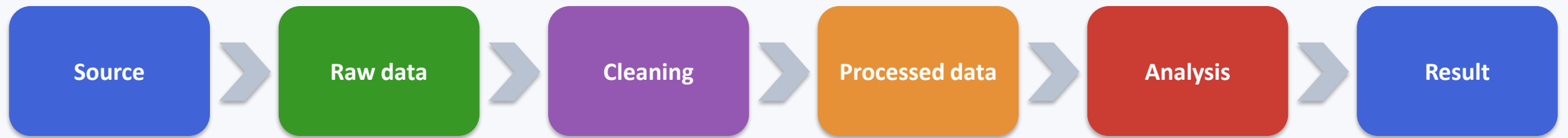
Context and documentation support future use.

Data provenance: the story of the data

Where did this value come from, and what happened to it?

- Source: system, instrument, survey, registry, or collaborator.
- Collection: when, where, by whom, and under what protocol.
- Transformation: cleaning, recoding, exclusions, derivations, linkage.
- Version: which snapshot or release was used.
- Responsibility: who performed or approved each step.

Provenance is a chain of evidence



Metadata: data about the data

Metadata provide the context needed to interpret a dataset.

Structural

File format • table relationships • variable names • data types

Descriptive

Study • population • collection dates • units • coding schemes

Administrative

Access conditions • version • ownership • license/DUA • provenance

A data dictionary is variable-level documentation

Variable	Meaning	Type	Units / coding	Missing
patient_id	Study identifier	string	not applicable	none expected
age_years	Age at admission	integer	years	NA
readmit_30d	Readmission within 30 days	binary	0=no; 1=yes	NA

Variable names are not definitions

Never force the analyst to guess.

Weak documentation

sex = 1/2
visit = 0/1
gluc = numeric
status = 1/2/3

Better documentation

sex_at_birth: 1=female, 2=male
visit: 0=baseline, 1=follow-up
fasting_glucose_mg_dl: mg/dL
diabetes_status: explicit coding

Derived variables need provenance too

A formula is part of the data lineage.

Example: BMI

$\text{BMI} = \text{weight_kg} / \text{height_m}^2$

Document source variables, units, formula, rounding, and handling of impossible values.

Example: 30-day readmission

Define index encounter, eligible discharge, time window, transfers, deaths, and what counts as a readmission.

README: the front door to the project

- What is this project and what question does it address?
- Where did the data come from, and what access restrictions apply?
- What is the directory structure?
- How are raw data transformed into processed data?
- What code should be run, and in what order?
- Who should be contacted for questions?

Document transformations, not just the final dataset

01 Import

Read source file without changing it.

02 Validate

Check types, ranges, duplicates, missingness.

03 Clean

Standardize dates/units; resolve documented errors.

04 Derive

Create analysis variables using explicit rules.

05 Export

Write a versioned processed dataset.

Raw data should be treated as immutable

Preserve the original evidence.

Do

Keep an untouched source copy • read raw data • write cleaned output to a new location • version code

Avoid

Opening the only raw file in Excel and overwriting values • undocumented manual edits • replacing source files

Git helps with provenance — but Git is not enough

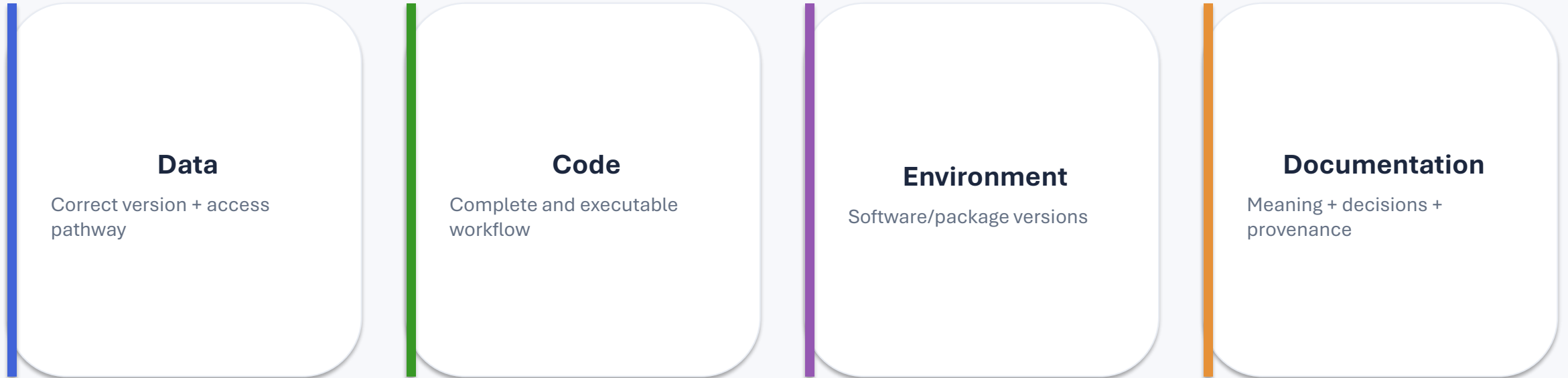
Git can record

Who changed code • when • what lines changed • commit history • branches and merges

Documentation must explain

Why the change was made • what a variable means • scientific rationale • data-access conditions

Reproducibility requires more than code



Documentation makes FAIR practical

Findable

Descriptive metadata • stable names • identifiers

Accessible

Access instructions • governance • contact/process

Interoperable

Standard formats • units • vocabularies

Reusable

Provenance • dictionary • license/conditions • processing notes

Hands-on lab: make the dataset understandable

Your task is to turn an undocumented dataset into a traceable research asset.

- Inspect a small synthetic biomedical dataset.
- Write/refine the research question and evaluate it with FINER.
- Create a data dictionary for the analysis variables.
- Document source and provenance.
- Write a README describing structure and workflow.
- Record at least three cleaning/derivation decisions and justify them.

Documentation quality checklist

- Could a new analyst identify the correct dataset and version?
- Could they understand every variable used in the analysis?
- Could they reconstruct each major transformation?
- Could they distinguish raw, processed, and derived data?
- Could they determine what access or sharing restrictions apply?
- Could they rerun the workflow without asking the original analyst what they meant?

Week 4 takeaways

- FINER evaluates whether a research question is feasible, interesting, novel, ethical, and relevant.
- FAIR guides stewardship: findable, accessible, interoperable, reusable.
- Provenance records where data came from and what happened to them.
- Data dictionaries explain variables; READMEs explain projects and workflows.
- Reproducibility depends on preserving both the computation and the context.

Optional statistical appendix: power analysis

A closer look at one component of feasibility.

Core idea

Power analysis connects the effect you want to detect, the variability in the data, the significance threshold, and the sample size.

Four quantities are linked in a power analysis

Effect size

How large a difference or association matters?

Sample size

How many independent units are observed?

α

Probability threshold for a Type I error; often 0.05.

Power

Probability of detecting the specified effect if it exists; often 80% or 90%.

What increases power?

Larger sample size

More information usually reduces uncertainty.

Larger true effect

A stronger signal is easier to distinguish from noise.

Lower variability

Less noise makes the signal easier to detect.

Worked example: comparing two independent means

Suppose we want 80% power, two-sided $\alpha = 0.05$, to detect a standardized mean difference $d = 0.50$.

Planning inputs

Effect size $d = 0.50$
 $\alpha = 0.05$
Power = 0.80
Two independent groups

Approximate result

≈ 64 participants per group
 ≈ 128 total

Then inflate for expected attrition/missingness.

A useful approximation for two equal groups

For a two-sided comparison of means with standardized effect d :

$$n \text{ per group} \approx 2 (z_{1-\alpha/2} + z_{1-\beta})^2 / d^2$$

- For $\alpha = 0.05$, $z_{1-\alpha/2} \approx 1.96$.
- For 80% power, $z_{1-\beta} \approx 0.84$.
- With $d = 0.50$: $n \approx 2(1.96 + 0.84)^2 / 0.50^2 \approx 63 \rightarrow$ about 64/group.

How to do a power analysis in practice

- Specify the primary outcome and statistical test/model.
- Choose a scientifically meaningful effect size.
- Estimate variability/event rate/correlation from prior evidence or pilot data.
- Set α and target power.
- Solve for sample size using appropriate software.
- Adjust for design features: attrition, clustering, repeated measures, multiplicity, unequal allocation.
- Document every assumption and run sensitivity analyses.

Software options for power and sample-size planning

R

power.t.test() for simple t-tests
pwr package for common
standardized-effect calculations
Specialized packages for complex
designs

G*Power

GUI-based calculations for many
common tests; useful for
teaching and quick planning.

Simulation

Often best for complex
longitudinal, clustered, survival,
or model-specific designs.

Power-analysis pitfalls

- Using an unrealistically large effect size to obtain a convenient sample size.
- Ignoring attrition, missingness, clustering, or unequal group sizes.
- Choosing the test after seeing the data.
- Treating “80% power” as proof that the study will succeed.
- Performing post-hoc “observed power” instead of reporting estimates and uncertainty.

One course-level message

A trustworthy analysis leaves a trail.

Question → Data → Decisions → Code → Result

FINER helps us ask whether the question is worth pursuing. FAIR, provenance, metadata, documentation, and version control help ensure that the path to the answer can be understood and reused.